

FollowUpAgent UC04 Report

May 12, 2026



#	Task	Status
1	Define system architecture (offline AI deployment)	✓
2	Identify hardware requirements for Datacore	✓
3	Design API contract with CRM team	✓

- Completed core architectural design for security-sensitive offline environment.
- Synchronized hardware specifications with Datacore infrastructure team.

Component	Minimum Requirement
CPU	12–16 vCPU
RAM	32 GB
GPU	1 × NVIDIA (RTX 3090/4090, A10G) ≥ 24GB VRAM
Storage	200 GB SSD
OS	Ubuntu 22.04 / 24.04
Network	Internal only (no internet)

Why 24GB+ VRAM?

Required to run Whisper and 7-8B LLMs (Llama-3/Qwen-2.5) with acceptable latency. CPU-only inference is not viable for production.

Step 1: CRM → AI Service

CRM sends candidate metadata and an internal audio URL.

POST /api/v1/candidates/analyze

```
{
  "ssn": "079099123456",
  "candidate_name": "Nguyen Van A",
  "audio_url": "http://192.168.1.200/files/interview.mp3"
}
```

Step 2: AI Service Response

The AI acknowledges the request and queues it for background processing.

Status: 202 Accepted

```
{
  "task_id": "TASK-a1b2c3d4",
  "status": "processing",
  "message": "Audio received; processing in background."
}
```

Step 3: AI Service → CRM Webhook

Once analysis is complete, the AI pushes the result back to the CRM.

POST /webhook/candidate-recommendation

```
{
  "task_id": "TASK-a1b2c3d4",
  "ssn": "079099123456",
  "status": "completed",
  "result": {
    "summary": "Solid technical skills. Clear communication.",
    "recommended_message": "HR will follow up within 2 days."
  }
}
```

- **Audio Access:** CRM must host audio on an internal URL reachable by the AI.
- **Reliability:** Recommend exponential-backoff retries for webhook failures.
- **Tracking:** CRM should store the `task_id` to match results to candidates.

#	Task	Target Date
1	Provision a small test server (VM) for the FastAPI service	Week 1 (May 19)
2	Produce Module Specification document (File responsibilities, interfaces, and API contracts)	Week 1 (May 19)
3	Set up a minimal FastAPI skeleton (Health-check + Routing)	Week 2 (May 26)
4	Validate core communication flows between CRM and AI	Week 2 (May 30)